Customer Order Analysis Project Report

Python Course-End Project

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Project Overview

This project analyzes a predefined set of e-commerce customer orders using Python lists, tuples, dictionaries, sets, loops, conditionals, list comprehensions, and sorting. The analysis classifies customers by spending, calculates revenue by product category, identifies purchasing patterns, and produces business recommendations.

Dataset and Data Structures

The program stores customer names in a list. Individual orders are stored as tuples containing customer name, product, price, and category. Dictionaries map customers to purchased products and products to categories. Sets are used to identify unique categories, unique products, multi-category customers, and customers common to multiple categories.

Customer Classification Results

Customer	Total Spent	Classification
Alice	$959.97	High-Value Buyer
Brian	$139.98	High-Value Buyer
Carla	$154.98	High-Value Buyer
David	$819.98	High-Value Buyer
Emma	$24.99	Low-Value Buyer
Frank	$339.98	High-Value Buyer
Grace	$174.97	High-Value Buyer
Henry	$64.99	Moderate Buyer

Product Category Analysis

Category	Revenue
Electronics	$2,009.95
Home Essentials	$379.94
Clothing	$289.95
Total	**$2,679.84**

Key Purchasing Patterns

Top three customers: Alice ($959.97), David ($819.98), and Frank ($339.98).

Electronics customers: Alice, Brian, David, Frank, and Grace.

Customers purchasing from multiple categories: Alice, Brian, David, Frank, and Grace.

Customers purchasing both Electronics and Clothing: Alice, Brian, David, and Grace.

Most frequently purchased products: Blender, Coffee Maker, Desk Lamp, Headphones, Jeans, and T-Shirt, each purchased twice.

Key Business Insights

1. Total revenue from all orders was $2,679.84.
2. Electronics was the most profitable category, producing $2,009.95 in revenue.
3. Six of eight customers were classified as high-value buyers.
4. Alice was the highest-spending customer with $959.97 in purchases.
5. Average customer spending was $334.98.
6. High-value customers should be targeted with loyalty offers and premium product recommendations.
7. Inventory planning should prioritize Electronics because it generated the largest share of revenue.
8. Cross-category customers are strong candidates for bundled offers and personalized promotions.

Challenges and Enhancements

Key challenges included selecting the correct data structure for each task, avoiding duplicate customers when generating category lists, implementing set intersections, and sorting customer totals correctly. The final program was enhanced with modular functions, formatted console tables, input-free repeatable analysis, reusable calculation functions, product-frequency analysis, and a dedicated business-insights section.

Conclusion

The project demonstrates how Python's built-in data structures can be used to convert raw order records into customer classifications, category-level sales analysis, purchasing-pattern insights, and practical business recommendations. The final program meets the assignment requirements and provides clear output suitable for management review.